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Wang et al.

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(54) **B7-H3 NANOBODY, PREPARATION METHOD AND USE THEREOF**

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C07K 16/28

(2006.01)

(52) **U.S. Cl.**

CPC **C07K 16/2827** (2013.01); **C07K 2317/22** (2013.01); **C07K 2317/569** (2013.01); **C07K 2317/76** (2013.01)

(58) **Field of Classification Search**

CPC combination set(s) only.

See application file for complete search history.

(56)

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International Search Report and Written Opinion mailed on Jan. 15, 2021, received for PCT Application PCT/CN2020/119893, Filed on Oct. 9, 2020, 19 pages including English Translation.

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(57)

ABSTRACT

The present invention provides a B7-H3 nanobody, the preparation method and use thereof. The B7-H3 nanobody comprises framework regions 1-4 (FR 1-4) and complementarity determining regions 1-3 (CDR 1-3), can specifically bind to B7-H3, and can be used for detecting B7-H3 molecules, and be used for the treatment of various malignant tumors with abnormal expression of B7-H3 molecule.

7 Claims, 1 Drawing Sheet

Specification includes a Sequence Listing.

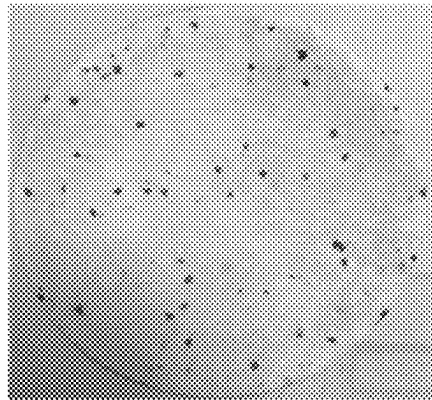


FIG.1

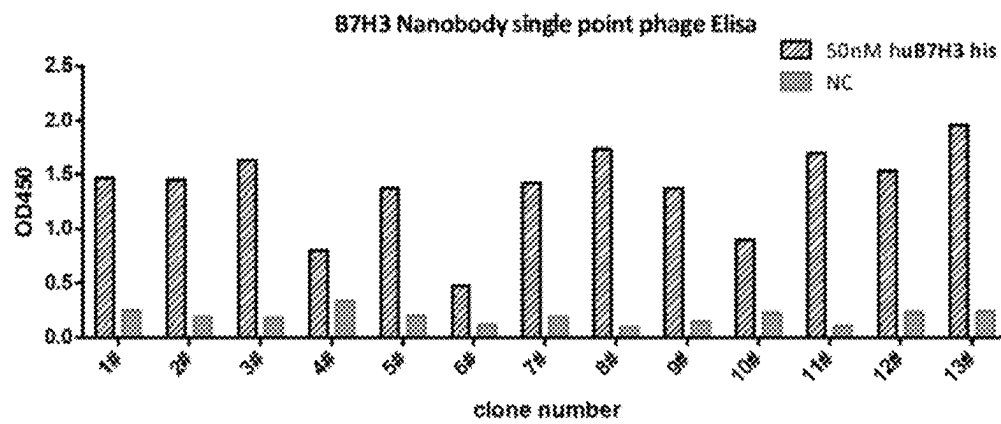


FIG.2

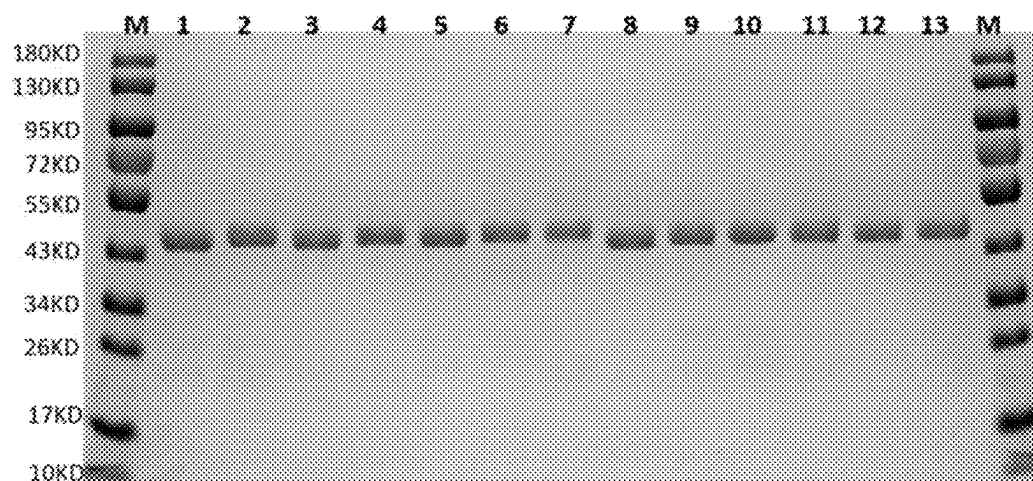


FIG.3

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B7-H3 NANOBODY, PREPARATION METHOD AND USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is based on PCT filing PCT/CN2020/119893, filed Oct. 9, 2020, which claims priority to Chinese Patent Applications No. 201910953967.4, filed on Oct. 9, 2019, the entire contents of each are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a nanobody against the antigenic epitope of a B7-H3 polypeptide molecule, a preparation method and use thereof.

BACKGROUND

B7-H3 (also known as CD276), a member of the B7 family, with the sequence similar to the extracellular domain of PD-L1 (B7-H1), is a phylogenetically conserved protein with various biological functions. B7-H3 is a type I transmembrane protein whose extracellular domain in mice consists of a pair of immunoglobulin variable domain (IgV) and immunoglobulin constant domain (IgC) (2IgB7-H3 isoform); in humans, it consists of two pairs of identical immunoglobulins produced by exon duplication (4IgB7-H3 isoform), which isoform is the predominant isoform expressed. B7-H3 is widely expressed at the mRNA level and can be detected in non-lymphoid organs such as liver, heart, prostate, and lymphoid organs such as spleen and thymus, however, its protein expression is strictly regulated. B7-H3 is structurally expressed on non-immune quiescent fibroblasts, endothelial cells, osteoblasts and amniotic fluid stem cells. After induction, B7-H3 can be expressed on immune cells, especially on antigen-presenting cells. Being co-cultured with regulatory T cells (Treg), dendritic cells can be induced to express B7-H3 protein with IFN- γ , lipopolysaccharide (LPS) or anti-CD40 stimulation in vitro. In addition, B7-H3 expression level can be up-regulated after monocytes and monocyte-derived DCs are stimulated by LPS or differentiated as induced by cytokines, respectively; B7-H3 protein can also be detected on NK cells, B cells, and a small number of T cells after PMA/ionomycin stimulation. Unlike restricted regulation in normal cells, B7-H3 is abnormally overexpressed in a variety of malignancies, including melanoma, leukemia, breast, prostate, ovarian, pancreatic, rectal and other cancers. High expression of B7-H3 is also associated with poor prognosis and poor clinical efficacy, and the development of antibody drugs targeting B7-H3 has become a promising tumor treatment strategy.

Nanobody technology is an antibody engineering revolution led by biomedical scientists on the basis of traditional antibodies by using molecular biology technology combined with the concept of nanoparticle sciences to develop the latest and smallest antibody molecules. It was originally found in camel blood by Belgian scientist Hamers, R. Ordinary antibody proteins are composed of two heavy chains and two light chains, while the nanoantibodies found in camel blood only have two heavy chains. Like traditional antibodies, these "heavy chain antibodies" can bind tightly to targets such as antigens, meanwhile, unlike single-chain antibodies, they are of a low aggregate tendency. Nanobody based on such "heavy chain antibody" is advantageous not only in a molecular weight that is only $\frac{1}{10}$ of that of

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traditional antibodies, but also in its more flexible chemical properties, good stability, high solubility, easy expression, high tumor tissue penetration and easy coupling to other molecules. Therefore, the application of nanobody technology to develop therapeutic antibodies against B7-H3 has great prospects.

SUMMARY OF THE INVENTION

In view of the above problems, on the one hand, the present invention provides a B7-H3 nanobody, which comprises framework regions (FR) and complementarity determining regions (CDR), the complementarity determining regions comprise a complementarity determining region 1 (CDR1), a complementarity determining region 2 (CDR2) and complementarity determining region 3 (CDR3), wherein,

the sequence of the complementarity determining region 1 (CDR1) is selected from any one of SEQ ID NO:5, SEQ ID NO:12, SEQ ID NO:19, SEQ ID NO:24, SEQ ID NO:29, SEQ ID NO:34, SEQ ID NO:41, SEQ ID NO:46, SEQ ID NO:51, SEQ ID NO:58, SEQ ID NO:63 and SEQ ID NO:68;

the sequence of the complementarity determining region 2 (CDR2) is selected from any one of SEQ ID NO:6, SEQ ID NO:13, SEQ ID NO:20, SEQ ID NO:25, SEQ ID NO:30, SEQ ID NO:35, SEQ ID NO:42, SEQ ID NO:47, SEQ ID NO:52, SEQ ID NO:59, SEQ ID NO:64, and SEQ ID NO:69; and

the sequence of the complementarity determining region 3 (CDR3) is selected from any one of SEQ ID NO:7, SEQ ID NO:14, SEQ ID NO:21, SEQ ID NO:26, SEQ ID NO:31, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:43, SEQ ID NO:48, SEQ ID NO:53, SEQ ID NO:60, SEQ ID NO:65, and SEQ ID NO:70.

In one embodiment, the B7-H3 nanobody comprises the following regions in order: a framework region 1 (FR1), a complementarity determining region 1 (CDR1), a framework region 2 (FR2), a complementarity determining region 2 (CDR2), a framework region 3 (FR3), a complementarity determining region 3 (CDR3) and a framework region 4 (FR4), wherein

the sequence of the framework region 1 (FR1) is selected from any one of SEQ ID NO:1, SEQ ID NO:8, SEQ ID NO:15, SEQ ID NO:22, SEQ ID NO:27, SEQ ID NO:32, SEQ ID NO:38, SEQ ID NO:44, SEQ ID NO:49, SEQ ID NO:54, SEQ ID NO:61 and SEQ ID NO:66;

the sequence of the complementarity determining region 1 (CDR1) is selected from any one of SEQ ID NO:5, SEQ ID NO:12, SEQ ID NO:19, SEQ ID NO:24, SEQ ID NO:29, SEQ ID NO:34, SEQ ID NO:41, SEQ ID NO:46, SEQ ID NO:51, SEQ ID NO:58, SEQ ID NO:63 and SEQ ID NO:68;

the sequence of the framework region 2 (FR2) is any one selected from any one of SEQ ID NO:2, SEQ ID NO:9, SEQ ID NO:16, SEQ ID NO:39, and SEQ ID NO:55;

the sequence of the complementarity determining region 2 (CDR2) is selected from any one of SEQ ID NO:6, SEQ ID NO:13, SEQ ID NO:20, SEQ ID NO:25, SEQ ID NO:30, SEQ ID NO:35, SEQ ID NO:42, SEQ ID NO:47, SEQ ID NO:52, SEQ ID NO:59, SEQ ID NO:64 and SEQ ID NO:69;

the sequence of the framework region 3 (FR3) is selected from any one of SEQ ID NO:3, SEQ ID NO:10, SEQ ID NO:17, SEQ ID NO:23, SEQ ID NO:28, SEQ ID

In one embodiment, the B7-H3 nanobody comprises a complementarity determining region 1 (CDR1), a complementarity determining region 2 (CDR2) and a complementarity determining region 3 (CDR3), wherein,

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the sequence of the complementarity determining region
1 (CDR1) is SEQ ID NO: 63,
the sequence of the complementarity determining region
2 (CDR2) is SEQ ID NO: 64,
the sequence of the complementarity determining region
3 (CDR3) is SEQ ID NO: 65.

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In one embodiment, the B7-H3 nanobody comprises a complementarity determining region 1 (CDR1), a complementarity determining region 2 (CDR2) and a complementarity determining region 3 (CDR3), wherein,

the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 68,

the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 69,

the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:70.

In one embodiment, the B7-H3 nanobody has an amino acid sequence selected from any one of the following:

SEQ ID NO:71, SEQ ID NO:72, SEQ ID NO:73, SEQ ID NO:74, SEQ ID NO:75, SEQ ID NO:76, SEQ ID NO:77, SEQ ID NO:78, SEQ ID NO:79, SEQ ID NO:80, SEQ ID NO:81, SEQ ID NO:82 and SEQ ID NO:83.

In another aspect, the present invention provides a method for preparing the above B7-H3 nanobody, comprising the following steps:

- 1). Constructing B7-H3 molecular nanobody library;
- 2). Screening B7-H3 nanobodies;
- 3). Detecting single positive clones by immunoblotting;
- 4). Further verifying positive clones with phage Elisa; and
- 5). Expressing B7-H3 nanobodies in eukaryotic expression system and purifying them.

In another aspect, the present invention provides a DNA molecule encoding the above-mentioned B7-H3 nanobody.

In yet another aspect, the present invention provides an expression vector comprising the above-mentioned DNA molecule.

In yet another aspect, the present invention provides the use of the above-mentioned B7-H3 nanobody in the preparation of a B7-H3 molecule detection reagent.

In another aspect, the present invention provides the use of the above-mentioned B7-H3 nanobody in the preparation of a medicament for the treatment of malignant tumors with high expression level of B7-H3 molecule.

Technical Effects of the Present Invention

The B7-H3 nanobody of the present invention can specifically bind to B7-H3, therefore, can be used to detect B7-H3 molecules with high sensitivity, and is expected to be used as a therapeutic antibody for the treatment of various malignant tumors with high expression level of B7-H3 molecule.

DESCRIPTION OF FIGURES

FIG. 1 shows an output phage immunoblot in Example 3.

FIG. 2 shows the results of further verification of positive clones with phage Elisa in Example 4.

FIG. 3 is the SDS-PAGE picture of the purified 1-13 #nanobodies in Example 5.

DETAILED EMBODIMENTS

The specific preparation process of the nanobodies of the present application will be described below with reference to specific examples. However, those skilled in the art should understand that the scope of the present application is not intended to be limited by these examples, but includes various equivalent and modified forms made by those skilled in the art.

Complete/incomplete Freund's adjuvant (sigma, F5506), streptavidin beads (invitrogen, 65002), casein (Thermo,

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37528), PROTRAN BA 85 (Whatman, 10401116), microtiter plate (greiner, 650061)), goat anti-M13HRP (GE, 279421).

Example 1: Construction of Nanobody Library Against B7-H3 Molecular

- (1) 1 mg of the hu4IgB7-H3 antigen produced by protein expression was mixed with complete Freund's adjuvant in an equal volume and a Xinjiang dromedary was immunized therewith;
- (2) Starting from the second week, the dromedary was immunized for 7 times with a mixture of 1 mg of the hu4IgB7-H3 antigen and incomplete Freund's adjuvant in an equal volume to stimulate B cells to express antigen-specific nanobodies;
- (3) After 7 immunizations, 100 ml of the peripheral blood lymphocytes of dromedary was collected for extracting total RNA;
- (4) cDNA was obtained by Reverse transcription and amplified by nested PCR to obtain VHH fragments;
- (5) VHH single-stranded DNA was annealed into the phage vector using the kunkel reaction;
- (6) The kunkel product was transformed into electrotransformation competent DH108, and the B7-H3 nanobody library was constructed and the library size was determined. The library size was 1.07×10^9 .

Example 2: Screening Nanobodies Against B7-H3

- (1) Liquid phase panning:
 1. 10^8 pfu input phage was blocked with a final concentration of 0.5% casein blocking buffer and incubated for 1 h at room temperature;
 2. At the same time, 100 μ l streptavidin magnetic beads were washed three times with sterile PBS, the supernatant was discarded, and the resultant was blocked with 500 μ l of 1% casein blocking buffer for 1 h at room temperature;
 3. The blocking solution in step 2 was discarded, 1 ml of the blocked phage was added to the magnetic beads, mixed well, rotated and mixed for 15 minutes to allow binding; (negative screening)
 4. The supernatant was added into a new sterile EP tube, hu4IgB7-H3-biotin antigen with a final concentration of 100 nM was added thereto, and the resultant was rotated and mixed at room temperature for 2 h to perform binding;
 5. At the same time, 100 μ l streptavidin magnetic beads were washed three times with sterile PBS, the supernatant was discarded, and the resultant was blocked with 500 μ l of 1% casein blocking buffer for 1 h at room temperature;
 6. The blocking solution in step 5 was discarded, 1 ml of the phage-hu hu4IgB7-H3-biotin mixture was added to the magnetic beads, mixed well, rotated and mixed for 15 minutes to perform binding;
 7. The supernatant was discarded and the magnetic beads were washed 10 times with 1 $\times$ PBS, and collected;
 8. 400 μ l of freshly prepared elution buffer containing 100 mM Triethylamine was added to the magnetic beads obtained in step 7, rotated and eluted for 10 min;
 9. 400 μ l of the supernatant was transferred to a sterile EP tube, 200 μ l of pH6.4 Tris-HCl was added for neutralization, after the screening, the output phage was obtained.

(2) Solid phase panning:

1. An immunotube was coated with 1 ml of 100 nM hu-hu4IgB7-H3 antigen solution, and incubated at 4° C. overnight;
2. The next day, the uncoated immune tube was rinsed with 1×PBS for 3 times, spun dry, and blocked with 4 ml of 1% casein blocking buffer with rotation at room temperature for 1 h;
3. 5×10¹² pfu first-round input phage was blocked with a final concentration of 1% BSA, supplemented with 1×PBS to a final volume of 1 ml, and rotated at room temperature for 1 h for blocking;
4. The blocking solution in the antigen-uncoated immunotube was discarded, the blocked phage was put into an antigen-uncoated immunotube, rotated for 2 h for binding (negative screening);
5. At the same time, the coating solution in the antigen-coated immunotube was discarded, and the resultant was rinsed three times with 1×PBS, spun dry, and blocked with 4 ml of blocking buffer with rotation at room temperature for 1 hour;
6. The blocking solution in the antigen-coated immunotube was discarded, and the negatively screened phage was transferred to the immunotube, and rotated for 2 h for binding;
7. After discarding the supernatant, the immunotube was rinsed 10 times with 1×PBST, and spun dry;
8. 1 ml of freshly prepared elution buffer containing 100 mM Triethylamine was added to the immunotube and incubated at room temperature for 10 min;
9. 1 ml of eluate was transferred to a sterile EP tube, and neutralized with 500 µl pH6.4 Tris-HCl. After the screening, output phage was obtained.

Example 3: Detection of Single Positive Clones by Immunoblotting (McCafferty J, Griffiths A D, Winter G, et al. Phage Antibodies: Filamentous Phage Displaying Antibody Variable Domains[J]. Nature, 1990, 348(6301):552-554. Huse W, Sastry L, Iverson S, et al. Generation of a Large Combinatorial Library of the Immunoglobulin Repertoire in Phage Lambda[J]. Biotechnology, 1989, 246(4935):1275-1281.)

- (1) 2000pfu output phage was plated on each LB plate and incubated at 37° C. for 6-8 h;
- (2) Whatman 0.45 µm NC membrane was coated with 5 ml of 2 µg/ml anti-M13 antibody and incubated at room temperature for 25 h;
- (3) The coating solution was discarded, the NC membrane was blocked with 5 ml of 1% casein blocking buffer, and incubated at room temperature for 1 h;
- (4) The blocking solution was discarded, and the NC membrane was washed three times with 1×PBS, and dried in air;
- (5) The NC membrane was attached on the LB plate grown with phage, punched holes for positioning, and incubated at room temperature overnight;
- (6) The next day, the membrane was removed and placed in a new petri dish, washed 3 times with 1×PBS, and incubated in 100 nM hu4IgB7-H3-biotin for 1 h at room temperature;
- (7) The membrane was washed 8 times with 0.1% PBST, added with 1:1000 diluted Neutravidin-AP and incubated at room temperature for 30 min;
- (8) The membrane was washed 8 times with 0.1% PBST, added with 10 ml AP chromogenic substrate (100 µl

BCIP+100 µl NBT), and developed color at room temperature for 10-30 min;

- (9) The positive clones were picked out for plaque PCR and sequencing;
- (10) The Sequences were aligned using DNAMAN sequence alignment software, and clones with the same CDR1, CDR2, and CDR3 were considered as the same clone.

Example 4: Further Verification of Positive Clones with Phage ELISA

- (1) Amplification of phage: XL1blue *E. coli* was seeded at 1:100 in LB liquid medium (containing 10 µg/ml tetracycline), grown to OD₆₀₀=0.6; XI1blue were transferred into 96-well deep-well plates, 600 µl/well, added with 15 µl of the picked phage elution for amplification overnight. The resultant was centrifuged at 4000 rpm for 30 min, and the supernatant was used for phage Elisa;
- (2) Elisa detection:
 1. The hu4IgB7-H3 antigen was diluted with 1×PBS to 100 nM, the ELISA plate was coated with 50 µl of the diluted solution per well, and incubated overnight at 4°C;
 2. After discarding the coating solution, the ELISA plate was spun dry, and rinsed 3 times with 1×PBS;
 3. 200 µl of 1% casein blocking buffer was added to each well for blocking at room temperature for 1 hour;
 4. After discarding the blocking solution, each well was added with 50 µl of high-titer phage, and incubated at room temperature for 2 hours;
 5. The ELISA plate was rinsed 6 times with 0.1% PBST and spun dry;
 6. 1:5000 diluted goat anti-M13-HRP antibody was added at 50 µl/well, and incubated at room temperature for 1 h. The plate was washed 6 times with 0.1% PBST and spun dry;
 7. 50 µl of TMB color developing solution was added to each well for developing color at room temperature for 5 minutes. 50 µl of 2M H₂SO₄ was added to stop the reaction, and the OD₄₅₀ value was read;
 8. The sample wells having OD values 3 times greater than the control well were regarded as positive clones, and 13 positive clones were finally obtained. (The DNA sequences of the 13 nanobodies obtained are SEQ ID No.: 84-96, respectively.)

Example 5: Expression of the Nanobodies in Eukaryotic Expression System and Purification Thereof

- (1) The VHH fragments of different clones obtained by the previous sequencing analysis were cloned into the PINfuse eukaryotic expression vector;
- (2) After sequencing and verifying, the plasmid was extracted;
- (3) HEK293F suspension cells were cultured in Freestyle 293 expression medium, when the density reached 1×10⁶/ml, and the survival rate was >90%, they were ready for transfection;
- (4) The mass ratio of PEI:plasmid was 3:1, and transfection was carried out at the ratio of 1 µg plasmid/1 ml HEK293F cells;
- (5) On the 5th-6th day after transfection, cell supernatant was collected, purified with proteinA strain to obtain

high-purity antibody protein, and the elution buffer was replaced with PBS using an ultrafiltration column. (The

amino acid sequences of the 13 nanobodies obtained are SEQ ID No.: 71-83, respectively.)

SEQUENCE LISTING

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Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Val Ala His Arg
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<400> SEQUENCE: 2

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 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FR3

<400> SEQUENCE: 3

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Met Leu Ser Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys Ala Ala
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 <211> LENGTH: 11
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 <213> ORGANISM: artificial sequence
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<223> OTHER INFORMATION: CDR2

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Gly

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<220> FEATURE:
<223> OTHER INFORMATION: FR1

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Ser Leu Arg Leu Ser Cys Thr Ala Ser Gly Tyr Thr Arg Ser
20 25 30

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<220> FEATURE:
<223> OTHER INFORMATION: FR2

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<220> FEATURE:
<223> OTHER INFORMATION: FR3

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Cys Ala Ala
35

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<223> OTHER INFORMATION: FR4

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<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

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1 5

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<212> TYPE: PRT

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<220> FEATURE:

<223> OTHER INFORMATION: CDR3

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Phe Gly Tyr

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<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: FR1

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20 25 30

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<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: FR2

<400> SEQUENCE: 16

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1 5 10

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-continued

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<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 17

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1 5 10 15

Leu Asn Ser Leu Glu Thr Glu Asp Thr Ala Met Tyr Tyr Cys Ala Lys
 20 25 30

<210> SEQ ID NO 18
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR4

<400> SEQUENCE: 18

Arg Gly Gln Gly Thr Gln Val Thr Val Ser Ser
1 5 10

<210> SEQ ID NO 19
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 19

Asn Tyr Ala Met Ser
1 5

<210> SEQ ID NO 20
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 20

Gly Ile Asn Ser Asp Gly Gly Thr Tyr Tyr Ser Asn Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 21
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 21

Gly Asn Gly Pro Ser Trp Thr Leu Gly Ser Val Asp Val Asn Ser
1 5 10 15

<210> SEQ ID NO 22
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 22

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1 5 10 15

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<210> SEQ ID NO 28
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 28

Arg Phe Thr Ile Ser Gln Asp Ile Ala Glu Asn Thr Val Tyr Leu Arg
1 5 10 15

Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys Ala Ala
20 25 30

<210> SEQ ID NO 29
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 29

Arg Lys Trp Met Gly
1 5

<210> SEQ ID NO 30
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 30

Ala Ile Tyr Thr Ser Gly Arg Thr Ala Val Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 31
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 31

Leu Thr Gly Ala Ser Asp Ser Pro Leu Arg Pro Ser Glu Tyr Asn Tyr
1 5 10 15

<210> SEQ ID NO 32
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 32

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser
20 25 30

<210> SEQ ID NO 33
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 33

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Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Met Tyr Leu Gln
1 5 10 15

Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys Ala Ala
20 25 30

<210> SEQ ID NO 34
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 34

Ser Tyr Asp Met Ser
1 5

<210> SEQ ID NO 35
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 35

Ala Ile Asn Ser Gly Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 36
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 36

Asp Pro Gly Ser Ser Leu Ala Val Gly Ala Trp Leu Lys Ala Ala Asp
1 5 10 15

Pro Phe Tyr Gly Tyr
20

<210> SEQ ID NO 37
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 37

Thr Ser Gly Thr Val Gly Ser Met Val Arg Phe Ser Pro Val Gly Asp
1 5 10 15

Phe Gly Tyr

<210> SEQ ID NO 38
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 38

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

-continued

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser
 20 25 30

<210> SEQ ID NO 39
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FR2

<400> SEQUENCE: 39

Trp Val Arg Gln Ala Pro Gly Arg Gly Leu Glu Trp Val Ser
 1 5 10

<210> SEQ ID NO 40
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FR3

<400> SEQUENCE: 40

Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Met Leu Tyr Leu Gln
 1 5 10 15

Met Asp Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys Ala Thr
 20 25 30

<210> SEQ ID NO 41
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 41

Gly Val Asp Met Ser
 1 5

<210> SEQ ID NO 42
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 42

Gly Phe Gly Ser Gly Gly Glu Ser Pro Ser Tyr Ala Asp Ser Val Lys
 1 5 10 15

Gly

<210> SEQ ID NO 43
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 43

Arg Trp Ala Glu Gly Tyr
 1 5

<210> SEQ ID NO 44
 <211> LENGTH: 30
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:

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<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 44

Asp	Val	Gln	Leu	Gln	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly
1				5					10					15	

Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Leu	Phe	Ser
			20					25					30

<210> SEQ ID NO 45

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 45

Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Asn	Thr	Leu	Thr	Leu	Gln
1				5					10					15	

Leu	Asn	Ser	Leu	Lys	Thr	Glu	Asp	Thr	Ala	Met	Tyr	Tyr	Cys	Ala	Lys
			20					25					30		

<210> SEQ ID NO 46

<211> LENGTH: 5

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 46

Arg	Tyr	Ser	Met	Ser
1				5

<210> SEQ ID NO 47

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 47

Gly	Ile	Thr	Asn	Gly	Gly	Ser	Ile	Thr	Lys	Tyr	Ala	Asp	Ser	Val	Lys
1				5					10					15	

Gly

<210> SEQ ID NO 48

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 48

Gly	Ala	Gly	Glu	Tyr	Gly	Pro	Val	Ser	Asp	Tyr	Val	Phe	Ser	Leu
1			5					10						15

<210> SEQ ID NO 49

<211> LENGTH: 30

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 49

Asp	Val	Gln	Leu	Gln	Glu	Ser	Gly	Gly	Gly	Ser	Val	Gln	Ala	Gly	Gly
1				5					10					15	

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Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Tyr Thr Ile Gly
 20 25 30

<210> SEQ ID NO 50
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FR3

<400> SEQUENCE: 50

Arg Phe Ala Ile Ser Glu Asp Asn Ala Lys Asn Thr Val Tyr Leu His
 1 5 10 15
 Met Asn Leu Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys Ala Ala
 20 25 30

<210> SEQ ID NO 51
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 51

Ser Asn Tyr Met Gly
 1 5

<210> SEQ ID NO 52
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 52

Ala Ile Tyr Thr Gly Gly Gly Asn Thr Tyr Tyr Ala Asp Ser Val Lys
 1 5 10 15

Gly

<210> SEQ ID NO 53
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 53

Asp Leu Arg Leu Arg Val Ala Val Leu His Pro Tyr Gly Tyr Asn Tyr
 1 5 10 15

<210> SEQ ID NO 54
 <211> LENGTH: 30
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FR1

<400> SEQUENCE: 54

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Thr Phe Ser
 20 25 30

<210> SEQ ID NO 55

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<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR2

<400> SEQUENCE: 55

Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ser
1 5 10 15

<210> SEQ ID NO 56
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 56

Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr Leu Gln
1 5 10 15

Leu Asn Asn Leu Lys Thr Glu Asp Thr Ala Met Tyr Tyr Cys Thr Lys
20 25 30

<210> SEQ ID NO 57
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR4

<400> SEQUENCE: 57

Gly Gly Gln Gly Thr Gln Val Thr Val Ser Ser
1 5 10

<210> SEQ ID NO 58
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 58

Asn Tyr Ala Met
1

<210> SEQ ID NO 59
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 59

Arg Ile Ser Ser Ser Gly Ile Thr Tyr Tyr Ala Asp Ser Val Lys Gly
1 5 10 15

<210> SEQ ID NO 60
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 60

Ser Ser Glu Ala Pro
1 5

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<210> SEQ ID NO 61
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 61

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ala Ile Tyr Arg
20 25 30

<210> SEQ ID NO 62
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 62

Arg Phe Thr Ile Ser Gln Asp Asn Ala Lys Asn Thr Val Tyr Leu Gln
1 5 10 15

Met Asn Asp Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys Ala Ala
20 25 30

<210> SEQ ID NO 63
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 63

Gly Asn Tyr Met Gly
1 5

<210> SEQ ID NO 64
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 64

Ala Ile Ala Thr Asn Gly Arg Thr Thr Tyr Tyr Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 65
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 65

Asp Asn Ser Trp Ser Cys Thr Val Val Gly Gly Leu Leu Ala Pro Ala
1 5 10 15

Tyr Asn Tyr

<210> SEQ ID NO 66
<211> LENGTH: 30
<212> TYPE: PRT

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<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR1

<400> SEQUENCE: 66

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Arg Phe Thr Tyr Ser
20 25 30

<210> SEQ ID NO 67
<211> LENGTH: 32
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: FR3

<400> SEQUENCE: 67

Arg Phe Thr Ile Ser Leu Asp Ile Ala Lys Lys Thr Leu Ser Leu Gln
1 5 10 15

Met Asp Ser Leu Lys Ser Glu Asp Thr Ala Met Tyr Tyr Cys Ala Ala
20 25 30

<210> SEQ ID NO 68
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR1

<400> SEQUENCE: 68

Trp Tyr Cys Met Gly
1 5

<210> SEQ ID NO 69
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR2

<400> SEQUENCE: 69

Ala Ile Cys Thr Thr Pro Ser Arg Thr Tyr Ala Ala Asp Ser Val Lys
1 5 10 15

Gly

<210> SEQ ID NO 70
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3

<400> SEQUENCE: 70

Asp Thr Arg Gly Ala Ile Gly Phe Phe Lys Gly Ser Gly Pro Ala Ile
1 5 10 15

Pro Tyr

<210> SEQ ID NO 71
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 1# nanobody

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<400> SEQUENCE: 71

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Val Ala His Arg Arg Tyr
20           25           30
Tyr Met Ser Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
35           40           45
Ala Ser Ile Tyr Ile Gly Ser Gly Ser Thr Asp Tyr Val Asp Ser Val
50           55           60
Lys Gly Arg Phe Thr Ile Ser Gln Asp Asn Ala Asn Asp Met Val Tyr
65           70           75           80
Leu Gln Met Leu Ser Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys
85           90           95
Ala Ala Gly Leu Tyr Gly Gly Arg Trp Asp Arg Ala Gln Tyr Lys Tyr
100          105          110
Trp Gly Gln Gly Thr Gln Val Thr Val Ser Ser
115          120

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<210> SEQ ID NO 72

<211> LENGTH: 131

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: 2# nanobody

<400> SEQUENCE: 72

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Thr Ala Ser Gly Tyr Thr Arg Ser Thr Ile
20           25           30
Trp Ile Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
35           40           45
Ala Ala Ile Tyr Ile Gly Ser Gly Ala Thr Tyr Tyr Val Asp Ser Val
50           55           60
Lys Gly Arg Phe Thr Ile Ser Gln Asp Asn Ala Lys Asn Ala Lys Asn
65           70           75           80
Thr Val Tyr Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Met
85           90           95
Tyr Tyr Cys Ala Ala Thr Gly Gly Thr Val Gly Ser Met Val Arg Phe
100          105          110
Ser Pro Val Gly Asp Phe Gly Tyr Trp Gly Gln Gly Thr Gln Val Thr
115          120          125
Val Ser Ser
130

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<210> SEQ ID NO 73

<211> LENGTH: 123

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: 3# nanobody

<400> SEQUENCE: 73

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Asn Tyr
20           25           30

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-continued

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Gly Ile Asn Ser Asp Gly Gly Thr Tyr Tyr Ser Asn Ser Val Lys
 50 55 60

Gly Arg Ser Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr Leu
 65 70 75 80

Gln Leu Asn Ser Leu Glu Thr Glu Asp Thr Ala Met Tyr Tyr Cys Ala
 85 90 95

Lys Gly Asn Gly Pro Ser Trp Thr Leu Gly Ser Val Asp Val Asn Ser
 100 105 110

Arg Gly Gln Gly Thr Gln Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 74
 <211> LENGTH: 126
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 4# nanobody

<400> SEQUENCE: 74

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Asp Thr Tyr Arg Arg Asn
 20 25 30

Thr Met Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
 35 40 45

Ala Ala Ile Tyr Thr Gly Gly Thr Ser Thr Ala Tyr Ala Asp Ser Val
 50 55 60

Arg Gly Arg Phe Thr Ile Ser Gln Asp Ser Ala Lys Asn Thr Val Tyr
 65 70 75 80

Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Ile Tyr Tyr Cys
 85 90 95

Ala Ala Gly Ser Val Trp Ser Arg Tyr Thr Trp Ser Trp Pro Ser Thr
 100 105 110

Tyr Ser Tyr Trp Gly Gln Gly Thr Gln Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 75
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 5# nanobody

<400> SEQUENCE: 75

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Glu Tyr Ser Tyr Ser Arg Lys
 20 25 30

Trp Met Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
 35 40 45

Ala Ala Ile Tyr Thr Ser Gly Arg Thr Ala Val Ala Asp Ser Val Lys
 50 55 60

Gly Arg Phe Thr Ile Ser Gln Asp Ile Ala Glu Asn Thr Val Tyr Leu
 65 70 75 80

Arg Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys Ala
 85 90 95

-continued

Ala Leu Thr Gly Ala Ser Asp Ser Pro Leu Arg Pro Ser Glu Tyr Asn
 100 105 110

Tyr Trp Gly Gln Gly Thr Gln Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 76
 <211> LENGTH: 130
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 6# nanobody

<400> SEQUENCE: 76

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Asp Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ala Ile Asn Ser Gly Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Met Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys
 85 90 95
 Ala Ala Asp Pro Gly Ser Ser Leu Ala Val Gly Ala Trp Leu Lys Ala
 100 105 110
 Ala Asp Pro Phe Tyr Gly Tyr Trp Gly Gln Gly Thr Gln Val Thr Val
 115 120 125
 Ser Ser
 130

<210> SEQ ID NO 77
 <211> LENGTH: 131
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 7# nanobody

<400> SEQUENCE: 77

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Thr Ala Ser Gly Tyr Thr Arg Ser Thr Ile
 20 25 30
 Trp Ile Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
 35 40 45
 Ala Ala Ile Tyr Ile Gly Ser Gly Ala Thr Tyr Tyr Val Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Gln Asp Asn Ala Lys Asn Ala Lys Asn
 65 70 75 80
 Thr Val Tyr Leu Gln Met Asn Ser Leu Lys Pro Glu Asp Thr Ala Met
 85 90 95
 Tyr Tyr Cys Ala Ala Thr Ser Gly Thr Val Gly Ser Met Val Arg Phe
 100 105 110
 Ser Pro Val Gly Asp Phe Gly Tyr Trp Gly Gln Gly Thr Gln Val Thr
 115 120 125
 Val Ser Ser

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130

<210> SEQ ID NO 78
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 8# nanobody

<400> SEQUENCE: 78

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Gly Val
          20          25          30

Asp Met Ser Trp Val Arg Gln Ala Pro Gly Arg Gly Leu Glu Trp Val
          35          40          45

Ser Gly Phe Gly Ser Gly Gly Glu Ser Pro Ser Tyr Ala Asp Ser Val
          50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Met Leu Tyr
65          70          75          80

Leu Gln Met Asp Ser Leu Lys Pro Glu Asp Thr Ala Val Tyr Tyr Cys
          85          90          95

Ala Thr Arg Trp Ala Glu Gly Tyr Trp Gly Gln Gly Thr Gln Val Thr
          100          105          110

Val Ser Ser
          115

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<210> SEQ ID NO 79
 <211> LENGTH: 124
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 9# nanobody

<400> SEQUENCE: 79

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Leu Phe Ser Arg Tyr
          20          25          30

Ser Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45

Ser Gly Ile Thr Asn Gly Gly Ser Ile Thr Lys Tyr Ala Asp Ser Val
          50          55          60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Thr
65          70          75          80

Leu Gln Leu Asn Ser Leu Lys Thr Glu Asp Thr Ala Met Tyr Tyr Cys
          85          90          95

Ala Lys Gly Ala Gly Glu Tyr Gly Pro Val Ser Asp Tyr Val Phe Ser
          100          105          110

Leu Arg Gly Gln Gly Thr Gln Val Thr Val Ser Ser
          115          120

```

<210> SEQ ID NO 80
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 10# nanobody

<400> SEQUENCE: 80

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Tyr Thr Ile Gly Ser Asn
          20          25          30
Tyr Met Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
          35          40          45
Ala Ala Ile Tyr Thr Gly Gly Gly Asn Thr Tyr Tyr Ala Asp Ser Val
          50          55          60
Lys Gly Arg Phe Ala Ile Ser Glu Asp Asn Ala Lys Asn Thr Val Tyr
65          70          75          80
Leu His Met Asn Leu Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys
          85          90          95
Ala Ala Asp Leu Arg Leu Arg Val Ala Val Leu His Pro Tyr Gly Tyr
          100          105          110
Asn Tyr Trp Gly Gln Gly Thr Gln Val Thr Val Ser Ser
          115          120          125

```

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<210> SEQ ID NO 81
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 11# nanobody

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<400> SEQUENCE: 81

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Val Ser Gly Phe Thr Phe Ser Asn Tyr
          20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45
Ser Arg Ile Ser Ser Ser Gly Ile Thr Tyr Tyr Ala Asp Ser Val Lys
          50          55          60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr Leu
65          70          75          80
Gln Leu Asn Asn Leu Lys Thr Glu Asp Thr Ala Met Tyr Tyr Cys Thr
          85          90          95
Lys Ser Ser Glu Ala Pro Gly Gly Gln Gly Thr Gln Val Thr Val Ser
          100          105          110

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Ser

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<210> SEQ ID NO 82
<211> LENGTH: 128
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 12# nanobody

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<400> SEQUENCE: 82

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Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ala Ile Tyr Arg Gly Asn
          20          25          30
Tyr Met Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
          35          40          45
Ala Ala Ile Ala Thr Asn Gly Arg Thr Thr Tyr Tyr Ala Asp Ser Val
          50          55          60

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Lys Gly Arg Phe Thr Ile Ser Gln Asp Asn Ala Lys Asn Thr Val Tyr
65 70 75 80

Leu Gln Met Asn Asp Leu Lys Pro Glu Asp Thr Ala Met Tyr Tyr Cys
85 90 95

Ala Ala Asp Asn Ser Trp Ser Cys Thr Val Val Gly Gly Leu Leu Ala
100 105 110

Pro Ala Tyr Asn Tyr Trp Gly Gln Gly Thr Gln Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 83
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 13# nanobody

<400> SEQUENCE: 83

Asp Val Gln Leu Gln Glu Ser Gly Gly Gly Ser Val Gln Ala Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Arg Phe Thr Tyr Ser Trp Tyr
20 25 30

Cys Met Gly Trp Phe Arg Gln Ala Pro Gly Lys Glu Arg Glu Gly Val
35 40 45

Ala Ala Ile Cys Thr Thr Pro Ser Arg Thr Tyr Ala Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Leu Asp Ile Ala Lys Lys Thr Leu Ser
65 70 75 80

Leu Gln Met Asp Ser Leu Lys Ser Glu Asp Thr Ala Met Tyr Tyr Cys
85 90 95

Ala Ala Asp Thr Arg Gly Ala Ile Gly Phe Phe Lys Gly Ser Gly Pro
100 105 110

Ala Ile Pro Tyr Trp Gly Gln Gly Thr Gln Val Thr Val Ser Ser
115 120 125

<210> SEQ ID NO 84
<211> LENGTH: 369
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 1# nanobody

<400> SEQUENCE: 84

gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc	60
tcctgtgcag cctctggtgt cgcccacagg cgctactaca tgtcctggtt ccgccaggct	120
ccagggaagg agcgcgaggg ggtcgcatcc atttatattg gtagtggtag cacagactat	180
gtcgactccg tgaagggccg attcaccatc tcccaagaca acgccaacga catggtgtat	240
ctgcaaatgc tcagcctcaa acctgaggac actgccatgt actactgtgc ggcaggcctt	300
tacggttgga gatgggatcg ggcccagtat aagtactggg gccaggggac ccaggtcacc	360
gtctcctca	369

<210> SEQ ID NO 85
<211> LENGTH: 393
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 2# nanobody

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<400> SEQUENCE: 85

gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc	60
tcctgtacag cctcgggata caccgcagc accatttga taggctgggt ccgccaggct	120
ccaggaagg agcgcgaggg ggtgcagct atttatattg gtagtgccg gacatactat	180
gtggactccg tgaagggccg attcaccatc tccaagaca acgccaagaa cgccaagaac	240
acgggtgatc tgcagatgaa cagcctgaaa cctgaggaca ctgccatga ctactgtgcg	300
gcaacgggcg gtactgtagg aagtatggc cgattctccc cagtgggcga ctttggttat	360
tggggccagg ggaccaggc caccgtctcc tca	393

<210> SEQ ID NO 86

<211> LENGTH: 369

<212> TYPE: DNA

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: DNA of 3# nanobody

<400> SEQUENCE: 86

gatgtgcagc tgcaggagtc tggaggaggc ttggtgcagg ctgggggtc tctgagactc	60
tcctgtgcag cctctggatt caccttcaca aactatgcca tgagctgggt ccgccaggct	120
ccaggaagg gactcgagtg ggtctcaggt attaatagtg atggtggcac atactattca	180
aactccgtga agggccgaag caccatctcc agagacaacg ccaagaacac gctgtatctg	240
caattgaaca gcctggagac tgaggacacg gccatgtatt actgtgcaaa aggaaacggc	300
ccgtcttga ccctggggag cgtggacgtt aactcccggg gccaggggac ccaggtcacc	360
gtctcctca	369

<210> SEQ ID NO 87

<211> LENGTH: 378

<212> TYPE: DNA

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: DNA of 4# nanobody

<400> SEQUENCE: 87

gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc	60
tcctgtgcag cctctggaga cacttacagg cgcaacacca tgggctgggt ccgccaggct	120
ccgggaagg agcgcgaggg ggtgcagct atttatactg gtggtactag tacagcctat	180
gccgactccg tgaggggccc attcaccatc tccaagaca gcgccaagaa cacgggtgat	240
ctgcaaatga acagcctgaa acctgaggac actgccatat actactgtgc ggcaggctca	300
gtgtgggtccc gttatacatg gtctggccc agtacgtata gctactgggg ccaggggacc	360
caggtcaccg tctcctca	378

<210> SEQ ID NO 88

<211> LENGTH: 372

<212> TYPE: DNA

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: DNA of 5# nanobody

<400> SEQUENCE: 88

gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc	60
tcctgtgcag cctctgaata cagctacagt aggaagtga tgggctgggt ccgccaggct	120
ccaggaagg agcgcgaggg ggtgcagct atttatacta gtggtaggac agccgttgcc	180

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gactccgtga agggccgatt caccatctcc caagacatcg ccgagaacac ggtgtatctg	240
cgtatgaaca gcctgaaacc tgaggacact gccatgtact actgtgcggc actcacaggt	300
gctagcgact caccgctacg cccaagcgag tataactact ggggccaggg gaccaggtc	360
accgtctcct ca	372

<210> SEQ ID NO 89
 <211> LENGTH: 390
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: DNA of 6# nanobody

<400> SEQUENCE: 89

gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc	60
tcctgtgcag cctctggatt cacattcagt agctacgaca tgagctgggt ccgccaggct	120
ccagggaagg ggctcgagtg ggtctcagct attaatagtg gtggtggtag cacatactat	180
gcagactccg tgaaggccg attcaccatc tccagagaca acgccaagaa cactgatgat	240
ctgcaaatga acagcctgaa acctgaggac actgccatgt actactgtgc ggcagatcct	300
ggttcaagtt tggcgggtgg agcttggtg aaggcggcgg atcccttcta tggttactgg	360
ggccagggga cccaggtcac cgtctcctca	390

<210> SEQ ID NO 90
 <211> LENGTH: 393
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: DNA of 7# nanobody

<400> SEQUENCE: 90

gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc	60
tcctgtacag cctccgata caccgcagc accatttga taggtgggt ccgccaggct	120
ccagggaagg agcgcgaggg ggctcgagct atttatattg gtagtggcg gacatactat	180
gtggactccg tgaaggccg attcaccatc tccaagaca acgccaagaa cgccaagaac	240
acggtgtatc tgcagatgaa cagcctgaaa cctgaggaca ctgccaatga ctactgtgcg	300
gcaacgagcg gtactgtagg aagtatggc cgattctccc cagtggcgga ctttggttat	360
tggggccagg ggaccaggt caccgtctcc tca	393

<210> SEQ ID NO 91
 <211> LENGTH: 345
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: DNA of 8# nanobody

<400> SEQUENCE: 91

gatgtgcagc tgcaggagtc tggaggaggc ttggtgcagc ctgggggggc tctgagactc	60
tcctgtgcag cctctggatt cacattcagt ggcgtcgaca tgagctgggt ccgccaggct	120
ccaggggagg ggctcgagtg ggtctcaggg ttggttagtg gtggtgaaag cccatcatat	180
gcagactccg tgaaggccg attcaccatc tccagagaca acgccaagaa aatgctgtat	240
ctgcagatgg acagtttgaa acctgaggac acagccgtgt attactgcgc caccgcgtgg	300
gctgagggt actggggcca ggggacccag gtcaccgtct cctca	345

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<210> SEQ ID NO 92
<211> LENGTH: 372
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 9# nanobody

<400> SEQUENCE: 92
gatgtgcagc tgcaggagtc tggaggaggc ttggtgcagc ctgggggggc tctgagactc      60
tcctgtgcag cctctggatt cctcttcagt agatattcca tgagttgggt ccgccaggct      120
ccagggaagg gactcgagtg ggtctcaggt attacgaatg gtggtagcat tacaaagtat      180
gcagactccg tgaaggcccg attcaccatc tccagagaca acgccaagaa cacactgact      240
ctgcaactga acagcctgaa aactgaggac acggccatgt attactgtgc caaaggcgcg      300
ggggagtatg gccctgttag tgactatgtc ttctctctca ggggccaggg gaccagggtc      360
accgtctcct ca                                                              372

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<210> SEQ ID NO 93
<211> LENGTH: 375
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 10# nanobody

<400> SEQUENCE: 93
gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc      60
tcctgtgcag cctctggata caccatcggg agcaactaca tgggctgggt ccgccaggct      120
ccagggaagg agcgcgaggg ggtcgcagct atttatactg gtggtggtaa tacatactat      180
gccgactccg tgaaggcccg attcgccatc tccgaagaca acgccaagaa cacgggtgat      240
ctgcacatga acctcctgaa acctgaggac actgccatgt actactgtgc ggcagacttg      300
agactacggg ttgtgtact ccaccctgat gggataaact actggggcca ggggaccag      360
gtcacctgtc cctca                                                            375

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<210> SEQ ID NO 94
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 11# nanobody

<400> SEQUENCE: 94
gatgtgcagc tgcaggagtc tggaggaggc ttggtgcagc ctgggggggc tctgagactc      60
tcctgtgcag tctctggatt caccttcagt aactatgcca tgagctgggt ccgccaggct      120
ccagggaagg ggctcgagtg ggtctcacgt attagtagta gtggtattac atactatgca      180
gactccgtga agggccgatt caccatctcc agagacaacy ccaagaacac gctgtatcta      240
caattgaaca acctgaaaac tgaggacacg gccatgtatt actgtacaaa aagttccgag      300
gccccgggtg gccaggggac ccaggtcacc gtctctctca                            339

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<210> SEQ ID NO 95
<211> LENGTH: 384
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 12# nanobody

<400> SEQUENCE: 95

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gatgtgcagc tgcaggagtc tggaggaggc tcggtacagg ctggagggtc tctgagactc    60
tcctgtgcag cctctggagc catctacaga ggcaactaca tgggctggtt ccgccaggct    120
ccaggggaagg agcgcgaggg ggtcgcggct atcgctacta atggctgtac aacatactat    180
gccgactccg tgaagggcgc attcaccatc tcccaagaca acgccaagaa cacgggtgat    240
ctgcaaatga acgacctgaa acctgaggac actgccatgt actactgtgc ggcagacaac    300
tcttggtcct gtacgggtgt aggtgggcta ctggcgccctg cctataacta ctggggccag    360
gggacccagg tcaccgtctc ctca                                           384

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<210> SEQ ID NO 96
<211> LENGTH: 381
<212> TYPE: DNA
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: DNA of 13# nanobody

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<400> SEQUENCE: 96

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gatgtgcagc tgcaggagtc tggaggaggc tcggtgcagg ctggagggtc tctgagactc    60
tcctgtgcag cctctagatt tacctacagt tggtagtgca tgggctggtt ccgccaggct    120
ccggggaagg agcgcgaggg ggtcgcagct atttgacga caccgagccg cacatacgct    180
gccgactccg tgaagggtcg attcaccatc tccctagaca tcgccaagaa gacgtgtctc    240
ctgcaaatgg acagcctgaa atctgaggac actgccatgt actactgtgc ggcagataca    300
agaggagcta taggggtctt caagggcagt gggcccgcca tcccctactg gggccagggg    360
accaggtca ccgtctctc a                                           381

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The invention claimed is:

1. A B7-H3 a single domain antibody, which comprises framework regions (FR) and complementarity determining regions (CDR), the complementarity determining regions comprise a complementarity determining region 1 (CDR1), a complementarity determining region 2 (CDR2) and complementarity determining region 3 (CDR3), wherein, the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 5, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 6, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:7; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 12, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 13, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:14; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 19, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 20, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:21; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 24, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 25,

the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:26; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 29, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 30, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:31; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 34, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 35, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:36; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 12, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 13, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:37; or the sequence of the complementarity determining region 1 (CDR1) is SEQ ID NO: 41, the sequence of the complementarity determining region 2 (CDR2) is SEQ ID NO: 42, the sequence of the complementarity determining region 3 (CDR3) is SEQ ID NO:43;

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or
 the sequence of the complementarity determining region
 1 (CDR1) is SEQ ID NO: 46,
 the sequence of the complementarity determining region
 2 (CDR2) is SEQ ID NO: 47,
 the sequence of the complementarity determining region
 3 (CDR3) is SEQ ID NO: 48;

or
 the sequence of the complementarity determining region
 1 (CDR1) is SEQ ID NO: 51,
 the sequence of the complementarity determining region
 2 (CDR2) is SEQ ID NO: 52,
 the sequence of the complementarity determining region
 3 (CDR3) is SEQ ID NO: 53;

or
 the sequence of the complementarity determining region
 1 (CDR1) is SEQ ID NO: 58,
 the sequence of the complementarity determining region
 2 (CDR2) is SEQ ID NO: 59,
 the sequence of the complementarity determining region
 3 (CDR3) is SEQ ID NO: 60;

or
 the sequence of the complementarity determining region
 1 (CDR1) is SEQ ID NO: 63,
 the sequence of the complementarity determining region
 2 (CDR2) is SEQ ID NO: 64,
 the sequence of the complementarity determining region
 3 (CDR3) is SEQ ID NO: 65;

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or
 the sequence of the complementarity determining region
 1 (CDR1) is SEQ ID NO: 68,
 the sequence of the complementarity determining region
 2 (CDR2) is SEQ ID NO: 69,
 the sequence of the complementarity determining region
 3 (CDR3) is SEQ ID NO: 70.

2. The B7-H3 a single domain antibody according to claim 1, wherein, the B7-H3 a single domain antibody has an amino acid sequence selected from any one of the following:

SEQ ID NO: 71, SEQ ID NO: 72, SEQ ID NO: 73, SEQ ID NO: 74, SEQ ID NO: 75, SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, SEQ ID NO: 80, SEQ ID NO: 81, SEQ ID NO: 82 and SEQ ID NO: 83.

3. A DNA molecule encoding the B7-H3 a single domain antibody of claim 1.

4. An expression vector comprising the DNA molecule of claim 3.

5. A method of preparing a B7-H3 molecule detection reagent using the B7-H3 a single domain antibody of claim 1.

6. A DNA molecule encoding the B7-H3 a single domain antibody of claim 2.

7. A method of preparing a B7-H3 molecule detection reagent using the B7-H3 a single domain antibody of claim 2.

* * * * *